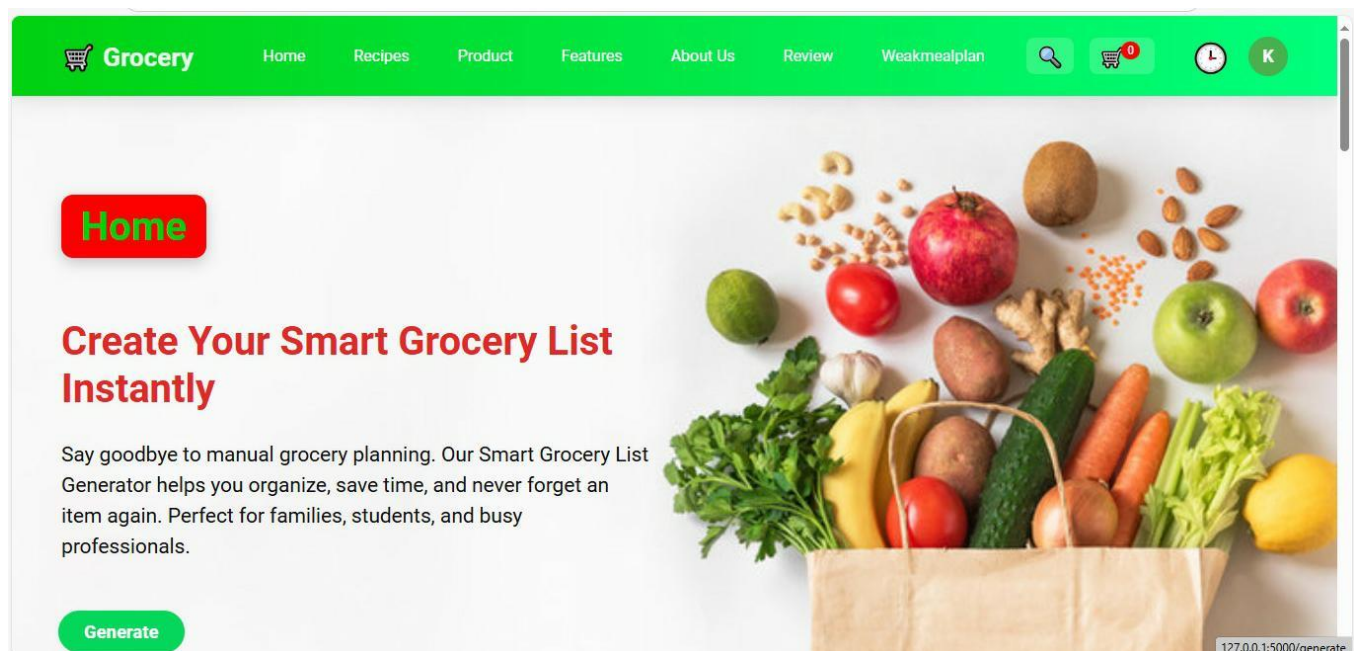


Smart Grocery List Generator

A web-based meal planning, grocery list and ordering platform

Degree	Bachelor of Computer Applications (Madurai Kamaraj University)
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Home page of the application after login

Project at a Glance

Planning meals and shopping for groceries is time-consuming, and it often ends in forgotten ingredients or overspending. The **Smart Grocery List Generator** solves this by letting a user pick a recipe or an entire week of meals and instantly receive a complete, quantity-adjusted ingredient list along with the total cost. From the same screen, the user can order the ingredients, so the whole journey from *planning* to *shopping* happens in a single, easy-to-use website.

1. Introduction

The Smart Grocery List Generator is a web application built with Python Flask on the backend and HTML, CSS and JavaScript on the frontend, connected to a database that stores users, recipes, ingredients, meal plans, search history and orders. Users sign up, log in, search for a recipe, enter the number of people they are cooking for, and the system calculates every ingredient quantity and its price automatically.

A standout feature is the **Weekly Meal Planner**. The user chooses breakfast, lunch and dinner for each of the seven days; the system merges the ingredients of all selected meals, adds up duplicate items, scales them by the number of people and produces one consolidated shopping list with a grand total. The user can then place the order directly from the platform.

2. Key Features

Feature	Description
Secure authentication	Signup, login and logout with hashed passwords and session handling.
Recipe search & ingredient list	Pick a recipe and number of people to get exact quantities, unit prices and total bill.
Weekly meal planner	7-day, 3-meal planner that produces a merged weekly shopping list with total cost.
Most searched recipes	Popular recipes with a "See Ingredients" view and cooking videos.
Product store & cart	Individual groceries can be added to a cart with a live cart-count badge.
Place order & bill	Checkout form, order confirmation and downloadable PDF bill.
Search history	Every recipe search of a logged-in user is stored and can be cleared.
Reviews	Customers can read reviews and submit their own without a page reload.

3. System Requirements

Hardware	Software
Processor: Intel i3 / AMD equivalent or higher RAM: 4 GB or more Storage: 500 GB Display: 15-inch or higher Internet connection	OS: Windows 10 / Linux / macOS Frontend: HTML5, CSS3, JavaScript Backend: Python Flask Database: SQLite / MySQL Browser: Chrome, Firefox, Edge

4. System Design

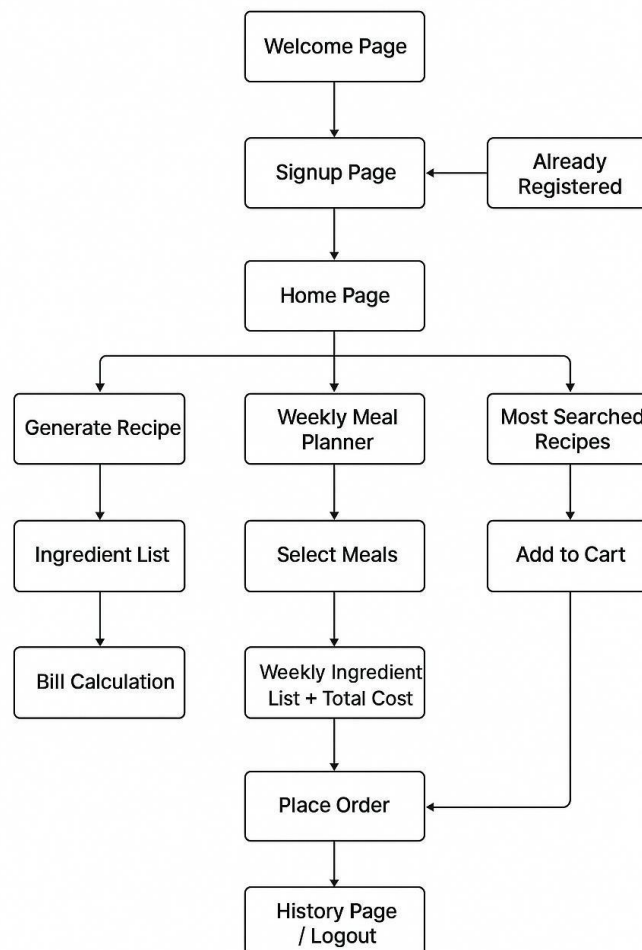
4.1 Module Design

The system is divided into seven modules, each responsible for one part of the experience:

- **Authentication Module** – registration, login and logout that protect user data.
- **Recipe Search Module** – finds recipes and generates the ingredient list for the selected dish.
- **Billing Module** – calculates the total cost for a recipe or a whole meal plan.
- **Meal Plan Module** – weekly 7-day planner to create, modify and view meal plans.
- **Place Order Module** – lets users order ingredients directly from recipes or plans.
- **UI Navigation Module** – navbar-based interface linking Home, Recipes, Product, Features, About Us, Review and Week Meal Plan.
- **Database Module** – stores users, recipes, ingredients, plans and orders with consistent, secure retrieval.

4.2 Application Flow

After the welcome page the user signs up (or logs in if already registered) and reaches the home page. From there three paths are available: **Generate Recipe** (ingredient list and bill), **Weekly Meal Planner** (select meals, weekly ingredient list and total cost) and **Most Searched Recipes** (add items to cart). Both the planner and the cart lead to **Place Order**, followed by the history page or logout.

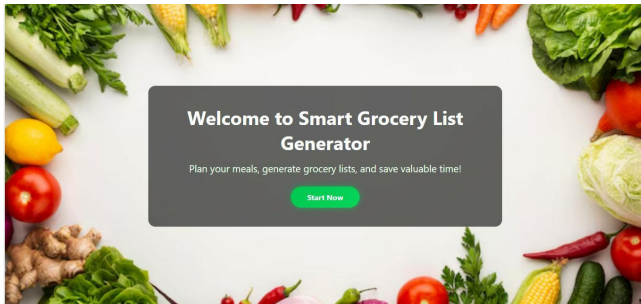


Flowchart of the Smart Grocery List Generator

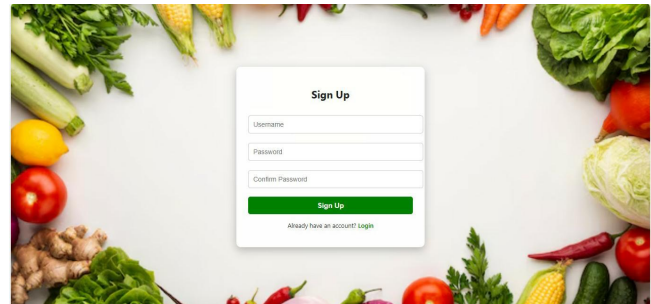
5. Implementation and Screenshots

5.1 User Access

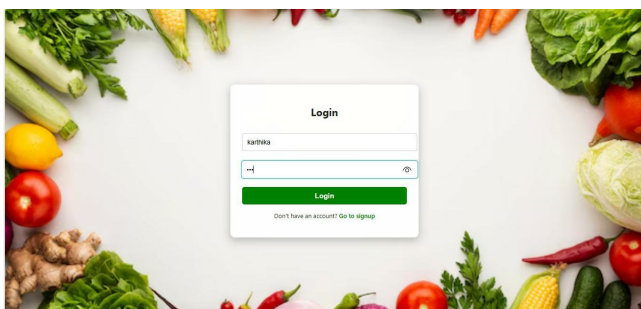
The application opens with a welcome page and a simple signup form. Returning users log in, which starts a session; the home page then shows the user's initial in a profile icon along with the Generate button and full navigation.



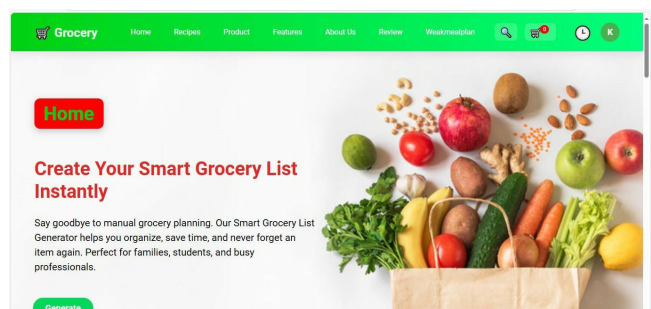
Welcome page



Signup page



Login page



Home page after login

5.2 Recipes and Ingredient List

Typing a recipe name and the number of people produces a table with each ingredient, scaled quantity, unit price and total. The popular recipes section shows ready cards such as Biryani, Vegetable Fried Rice, Pasta and Sambar, each with detailed ingredients and video guides.

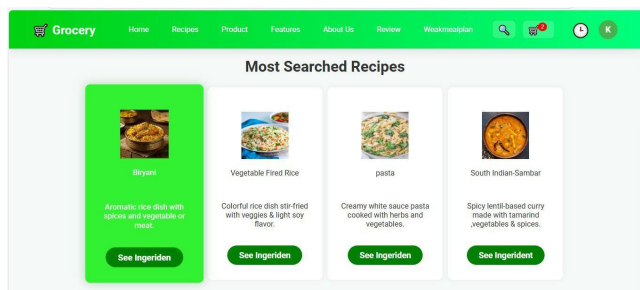
5.3 Generating a Grocery List

The screenshot shows a web interface for generating a grocery list for a 'Pasta' recipe. At the top, there is a search bar with 'Pasta' entered. Below it, a field for 'Enter Quantity (People):' has the value '4'. A green 'Generate' button is positioned below the quantity field. The main content area displays the 'Ingredients for Pasta' in a table format. The table has five columns: '#', 'Ingredient', 'Quantity', 'Unit Price', and 'Total'. It lists three items: Pasta (4 x 1 cup, ₹25.00, ₹100.00), Tomato sauce (4 x 1/2 cup, ₹10.00, ₹40.00), and Cheese (4 x 50g, ₹20.00, ₹80.00). Below the table, the 'Total Bill' is ₹220.00. At the bottom, there are three green buttons: 'Place Order', 'Week Plan', and 'Go to Home'.

#	Ingredient	Quantity	Unit Price	Total
1	Pasta	4 x 1 cup	₹25.00	₹100.00
2	Tomato sauce	4 x 1/2 cup	₹10.00	₹40.00
3	Cheese	4 x 50g	₹20.00	₹80.00

Total Bill: ₹220.00

Ingredient list and total bill for 4 people (Pasta: Rs. 220)



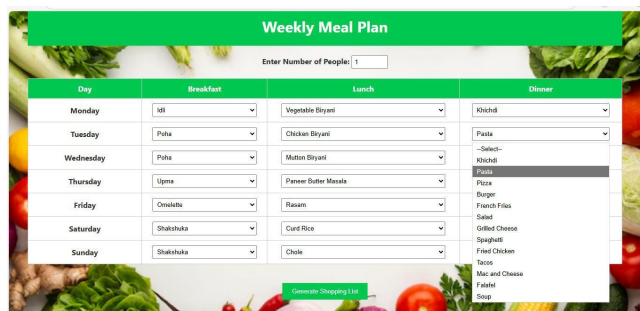
Most searched recipes



Ingredients and cooking video for Biryani

5.4 Weekly Meal Planner

The planner takes the number of people and a choice of breakfast, lunch and dinner for each day. When the shopping list is generated, ingredients that repeat across meals are combined into a single line with a merged quantity and a summed price. In the sample plan shown, the weekly grand total came to **Rs. 1066**.



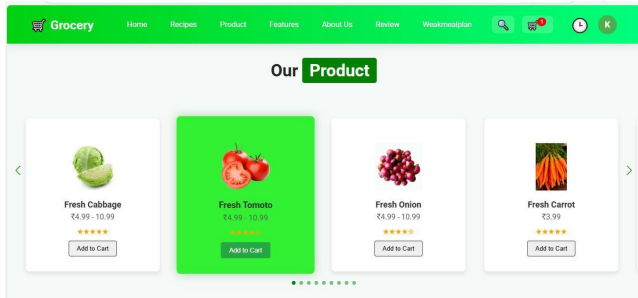
Meals selected for each day

The screenshot shows the 'Shopping List' table, which is a consolidated list of ingredients and their quantities, unit prices, and total prices. The table has four columns: 'Ingredient', 'Quantity', 'Unit Price', and 'Total Price'.

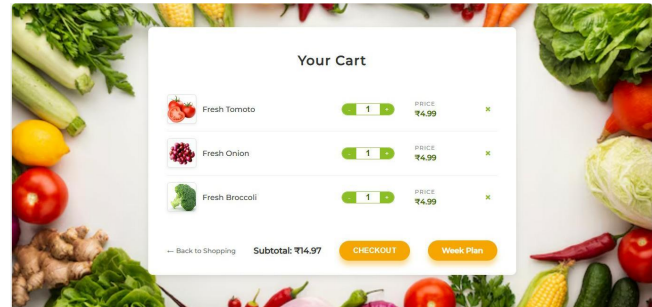
Ingredient	Quantity	Unit Price	Total Price
Idli rice	1 cup	₹15	₹15
Urad dal	0.25 cup	₹10	₹10
Salt	1 tsp + 0.10	₹1	₹1
Basmati rice	1 cup	₹30	₹30
Mixed vegetables	1 cup	₹20	₹20
Biryani masala	1 tbsp	₹10	₹10
Curd	1.5 cup	₹5	₹10
Rice	0.5 cup	₹10	₹10
Moong dal	0.5 cup	₹10	₹10
Ghee	3 tsp	₹5	₹15
Flattened rice	2 cup	₹10	₹20
Onion	2 chopped + 2 medium + 4	₹5	₹40
Mustard seeds	3 tsp	₹2	₹6
Green chilli	2	₹2	₹4

Consolidated weekly shopping list

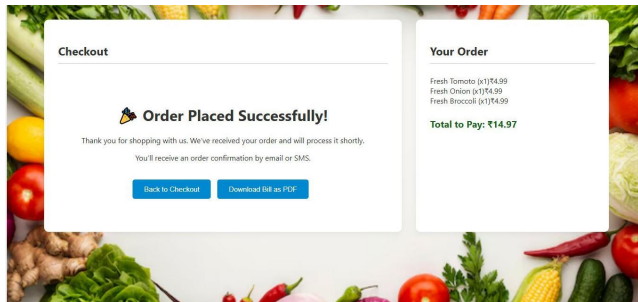
5.5 Products, Cart and Ordering



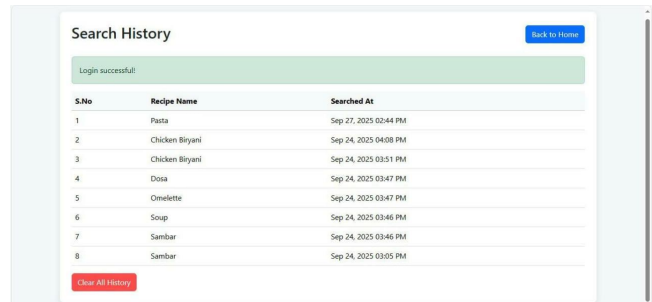
Product slider with Add to Cart



Cart with quantity controls and subtotal



Order placed successfully, with PDF bill



Recipe search history

5.6 Other Sections

The site also includes a **Features** section (Fresh and Organic, Free Delivery, Easy Payments) with detailed "Read More" pages, an **About Us** page, and a **Customer Review** area where new reviews are submitted without reloading the page.

6. Conclusion

The Smart Grocery List Generator brings recipe search, ingredient calculation, billing, weekly meal planning and direct ordering into one convenient platform. By combining Flask, a relational database and a responsive HTML/CSS/JavaScript interface, it saves users time, reduces missed ingredients and makes budgeting transparent. It is a practical example of how modern web technologies can simplify everyday household management.

References: Flask Documentation (flask.palletsprojects.com) • Python Documentation (docs.python.org) • W3Schools Tutorials (w3schools.com)